# Optimizing Data Retrieval Techniques

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# Overview

Advanced RAG and Retrieval Optimization

Retrieval System Optimization

Demos: Retrieval Process & Optimization

Relevance Scoring and Re-ranking

Demo: Implementing Re-ranking

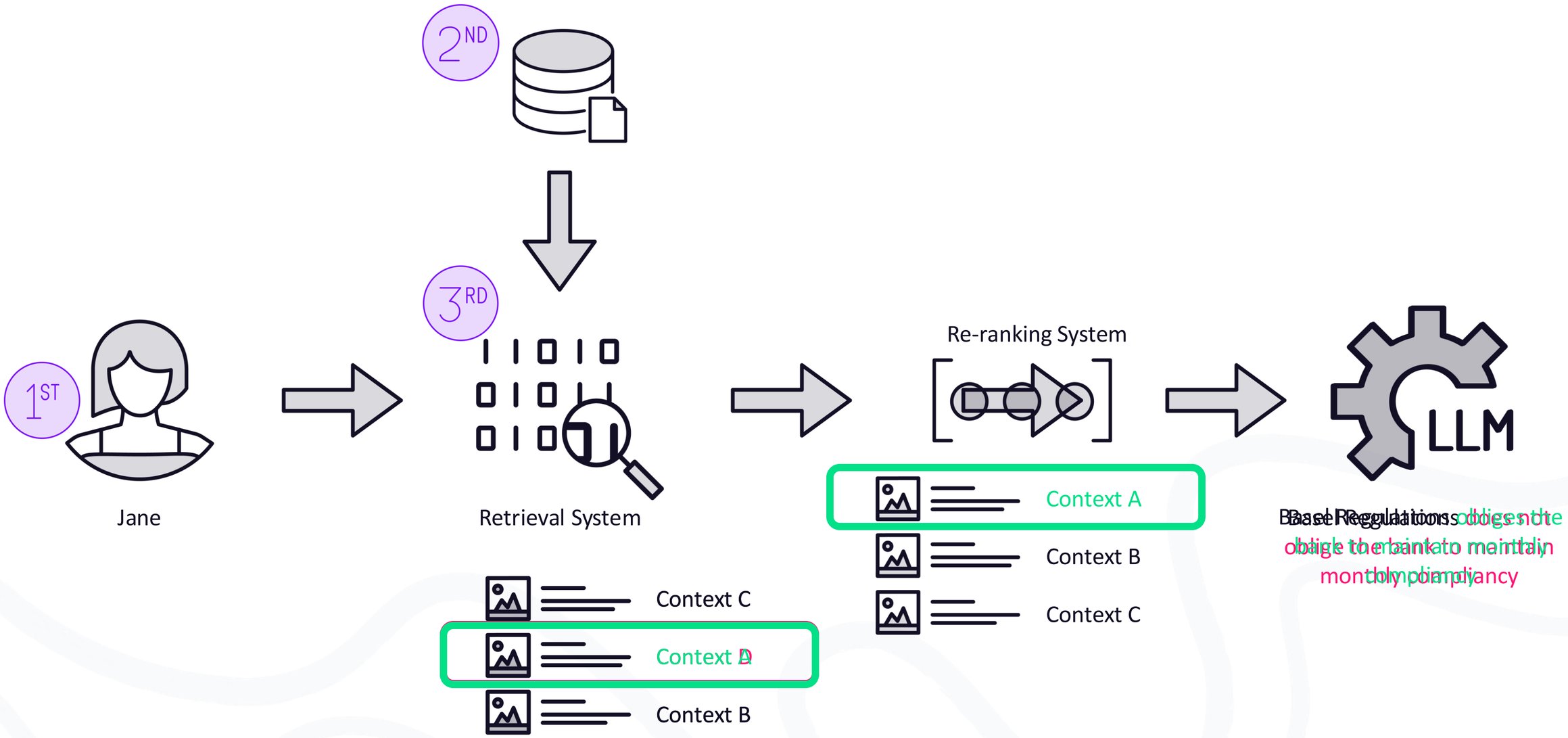
# Advanced RAG and Retrieval Optimization

# Naive RAG Versus Advanced RAG

Naive RAG includes indexing, retrieval, augmentation and generation

Advanced RAG is Naive RAG with enhanced retrieval quality and post-retrieval strategies

# Why Is Retrieval Optimization Important?



# Retrieval System Optimization

# Various Techniques for Optimizing Retrieval System

Vector Search

Fine-tuning embedding models for domain-specific contexts

Vector Search

Dynamic embeddings for context in which words are used

Hybrid Search

Combines vector search with keyword-based search

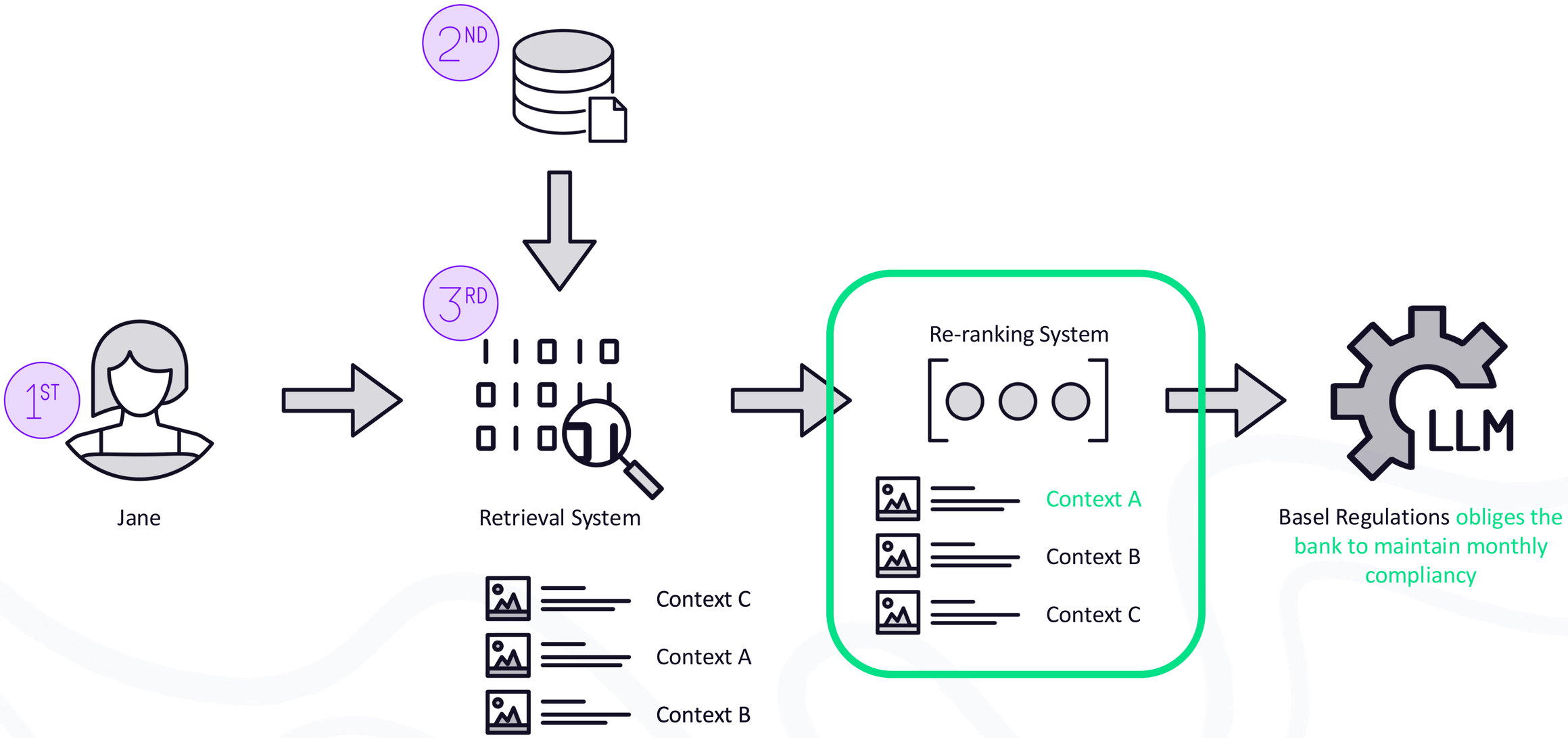
# Reserved for Demo

# Reserved for Demo

# Reserved for Demo

# Relevance Scoring and Re-ranking

# Relevance Scoring and Re-ranking



# Relevance Scoring and Re-ranking Methods

Basic Re-Ranking and Filtering

Advanced Re-Ranking with Machine Learning

# Reserved for Demo